SINAI UNIVERSITY
ARISH · KANTARA

CSC431---Robotics

Robotics Fundamentals

Fall 2025

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Outline

- From Last Lecture
- Robot Taxonomy
- Robot Applications
- Main Robot Components
- Summary

From Last Lecture

What Is A Robot?

- "a robot is an artificial physical agent that performs tasks by manipulating the physical world"

Robot Taxonomy

Robot Taxonomy

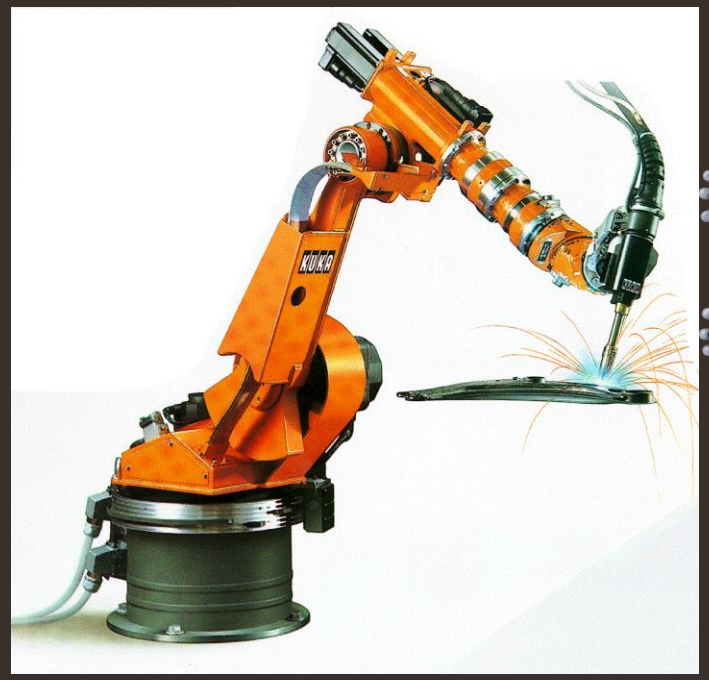
- The Japanese Industrial Robot Association gives the following classification of robots:
  - Class 1: Manual Handling Device
  - Class 2: Fixed Sequence Robot
  - Class 3: Variable Sequence Robot
  - Class 4: Playback Robot
  - Class 5: Numerical Control Robot
  - Class 6: Intelligent Robot

Robot Taxonomy

- We are primarily interested in intelligent robots and can break these down simply as follows:
  - Manipulator robots
  - Mobile robots
  - Humanoid robots

Manipulator Robots

- Physically anchored to their workplace
- Manipulator motion usually involves an entire chain of controllable joints, enabling such robots to place their effectors in any position within the workplace
- Manipulators are by far the most common type of industrial robots - a 2 billion dollar industry!



Mobile Robots

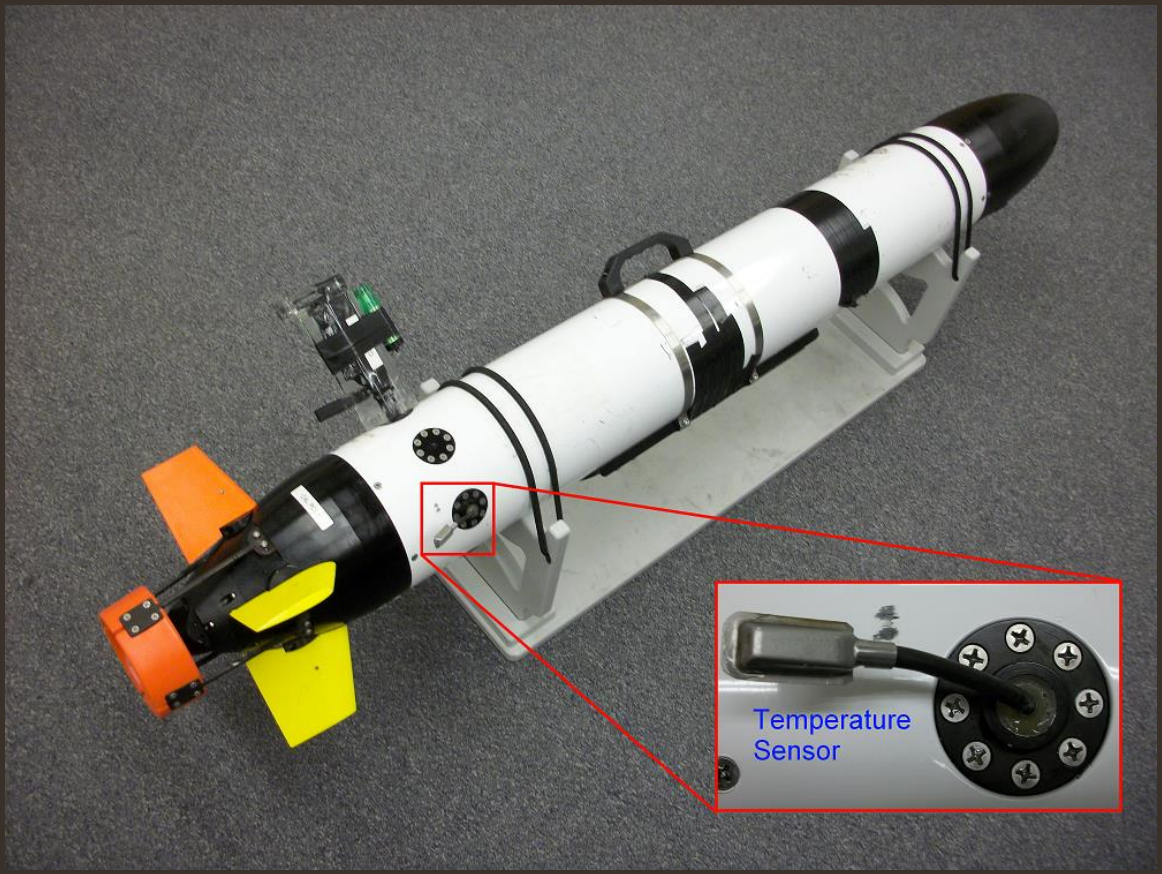
- We can divide mobile robots into the following categories:
  - Wheeled robots
  - Legged robots
  - Aerial robots (UAVs)
  - Others

Wheeled Robots

Legged Robots

Aerial Robots

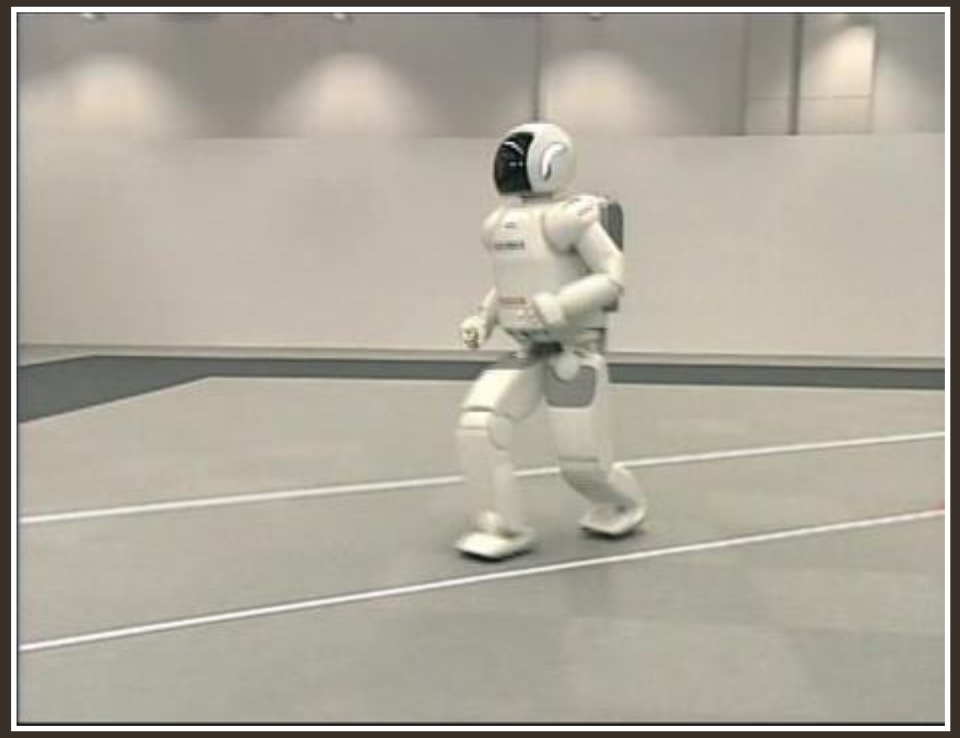
Other Mobile Robots



Autonomous Underwater Vehicle (AUV)

Humanoid Robots

- Hybrid robot: a mobile robot equipped with manipulators
- Hybrids can apply their effectors further a-field then anchored manipulators
- However, their task is made harder because they don't have the rigidity that the anchor provides.



Robot Applications

Robot Applications

- Indoor Structured Environments
  - Transportation
  - Customer support in museums, shops, etc.
  - Cleaning large buildings
  - Building surveillance
  - Research
  - Entertainment
- Outdoor Unstructured Environments
  - Space
  - Mining
  - Sewage tubes
  - Forest
  - Agriculture
  - Construction
  - Fire fighting
  - Military
  - Underwater
  - Aerial

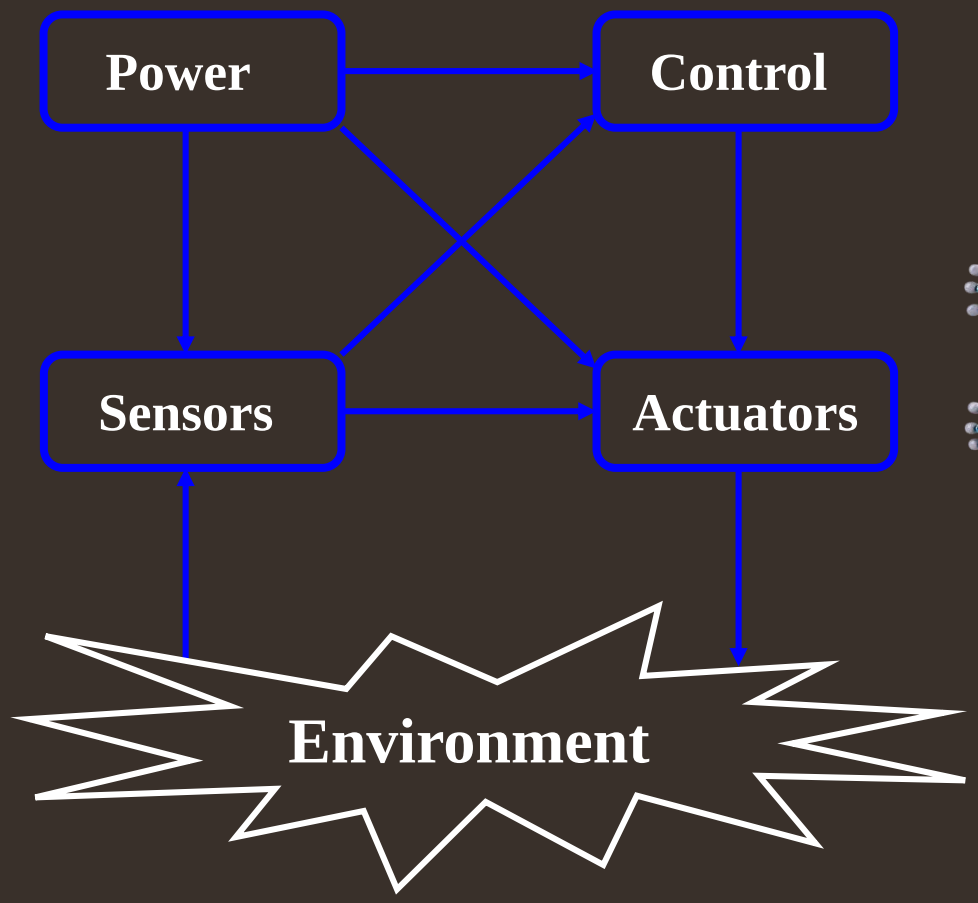
Advantages

- The major advantages of robots are:
  - Decreased labor costs
  - Increased precision and productivity
  - Increased flexibility compared with specialised machines
  - Robots can perform dull, repetitive jobs
  - Robots can operate in hazardous environments

Main Robot Components

Main Robot Components

- The major components of a mobile robot are:
  - Power supply
  - Sensors
  - Control
  - Actuators



Main Robot Components

- **Power Supply**: the *power plant and associated power transmission system*, such as a battery with associated equipment, or a power supply unit which converts electricity from the mains to appropriate power requirements

- **Sensors**: Including lasers, cameras, sonar and microphones, sensors enable robots to collect information about their own state and the state of their environment

Main Robot Components

- **Actuators**: devices which transduce power into kinetic energy or movement
  - Types of actuators include:
    - Electric motors
    - Pneumatic actuation (using compressed gas)
    - Hydraulic actuation (using pressurized fluids)
    - Robotic end effectors (e.g. grippers)

- **Control Subsystem**: responsible for evaluating the information collected and processed by sensors and for planning actions

Robot Control Mechanisms

- Robots may be controlled:
  - Directly by a human, such as remotely-controlled bomb-disposal robots, robotic arms, or shuttles
  - Autonomously according to their own decision making ability, provided by artificial intelligence

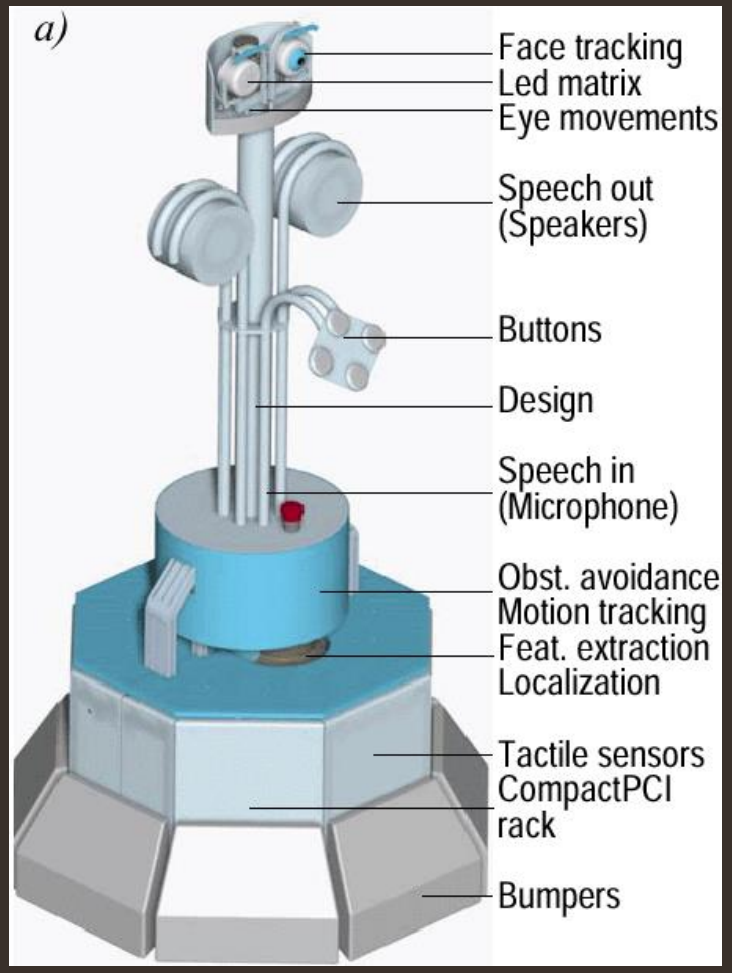
- Many robots fall in-between these extremes, being controlled by pre-programmed computers.

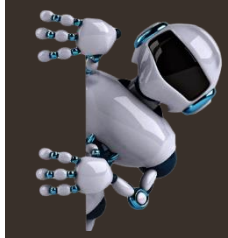
Mobile Robot Control

- **Tele-operated**
  - **Pioneer, robot that was designed to explore the Sarcophagus at Chernobyl**

Mobile Robot Control

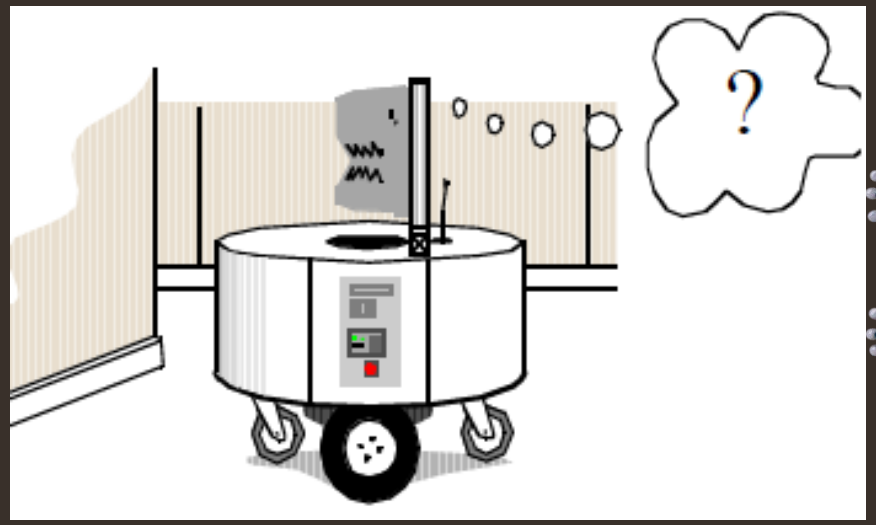
- Autonomous
  - EPFL Guide Robot
  - We are primarily interested in autonomous robots





Autonomous Mobile Robots

- The three key questions in Mobile Robotics
  - Where am I ?
  - Where am I going ?
  - How do I get there ?
  - How can I enhance my performance?
- To answer these questions the robot has to
  - have a model of the environment
  - perceive and analyze the environment
  - find its position within the environment
  - plan and execute the movement
  - Learn to select its behaviors
- This course will deal with Locomotion, Navigation, and Learning
  - (Perception, Localization, Planning and Learning)



Content of the Course

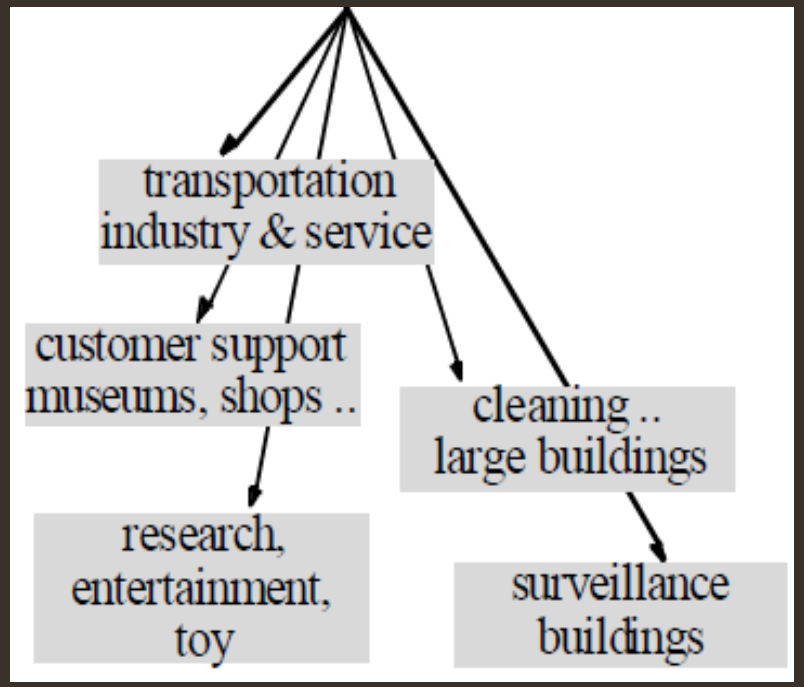
- 1. Introduction
- 2. Locomotion
- 3. Behaviour-based robotics
- 4. Mobile Robot Kinematics
- 5. Perception
- 6. Mobile Robot Localization
- 7. Planning and Navigation
- 8. Robot Learning

- Other Aspects of Autonomous Mobile Systems
- Applications

Applications of Mobile Robots

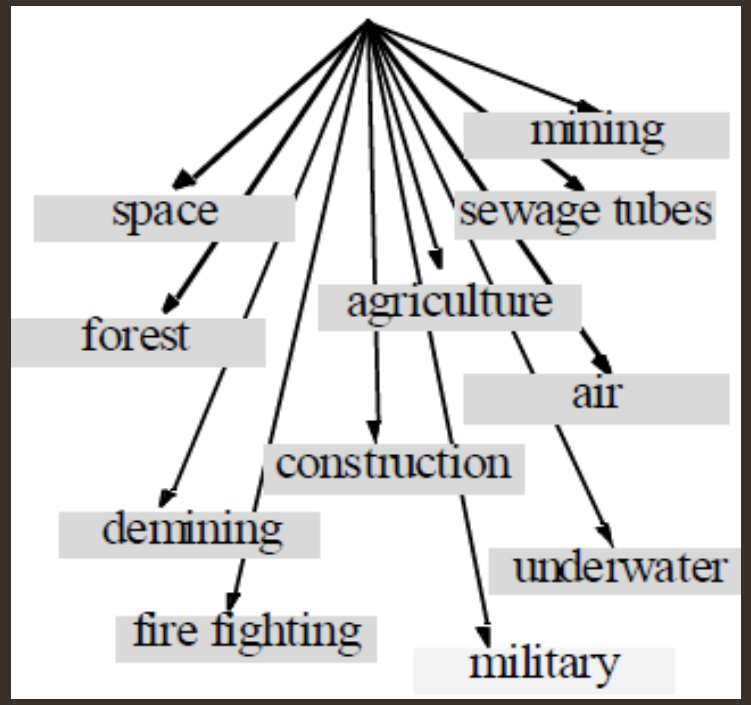
Indoor

Structured Environments

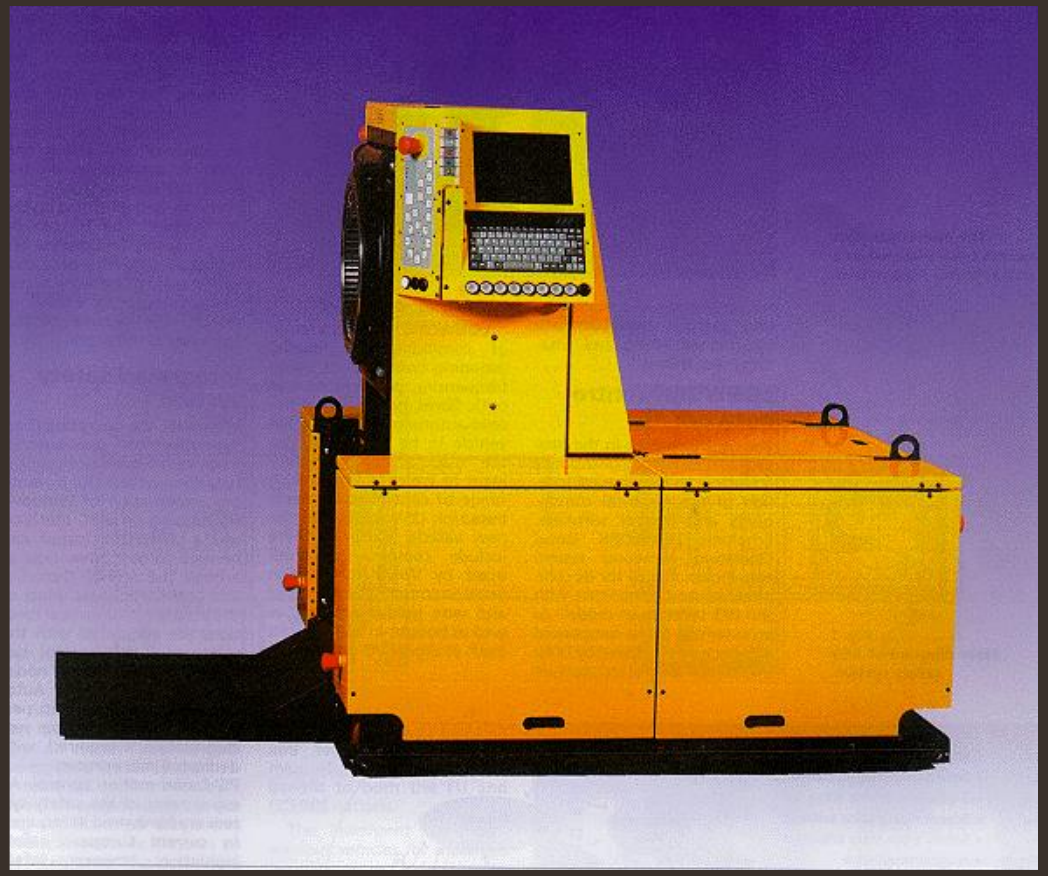


Outdoor

Unstructured Environments



Automatic Guided Vehicles



- Newest generation of Automatic Guided Vehicle of VOLVO used to transport motor blocks from on assembly station to an other. It is guided by an electrical wire installed in the floor but it is also able to leave the wire to avoid obstacles.

Helpmate

- HELPMATE is a mobile robot used in hospitals for transportation tasks. It has various on board sensors for autonomous navigation in the corridors. The main sensor for localization is a camera looking to the ceiling. It can detect the lamps on the ceiling as reference (landmark). http://www.ntplx.net/~helpmate/

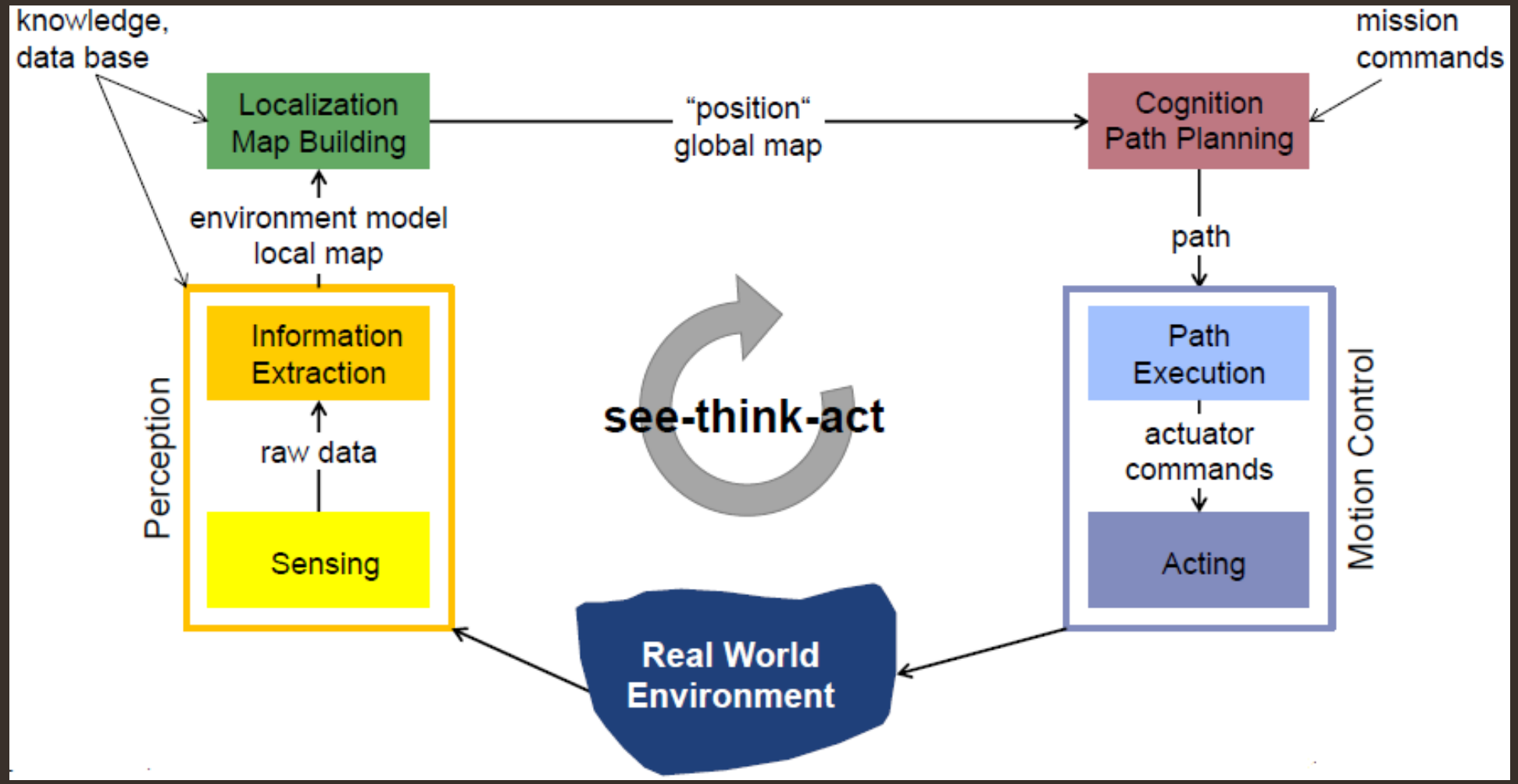
The Pioneer

- PIONEER I is a modular mobile robot offering various options like a gripper or an on board camera. It is equipped with a sophisticated navigation library developed at Stanford Research Institute (SRI).

The Khepera Robot

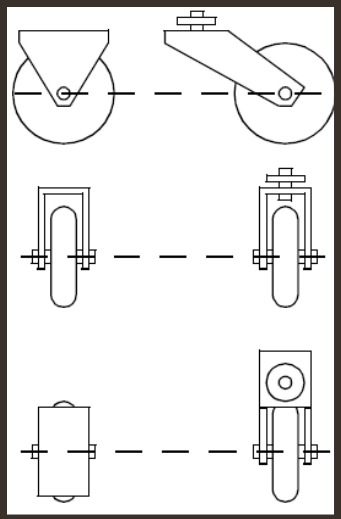
- KHEPERA is a small mobile robot for research and education. It is only about 60 mm in diameter. Additional modules with cameras, grippers and much more are available. More than 700 units were sold by the end of 1998).

Autonomous mobile robot | the see-think-act cycle



Motion Control | kinematics and motion control

- **Wheel types and its constraints**
  - Rolling constraint
  - no-sliding constraint (lateral)

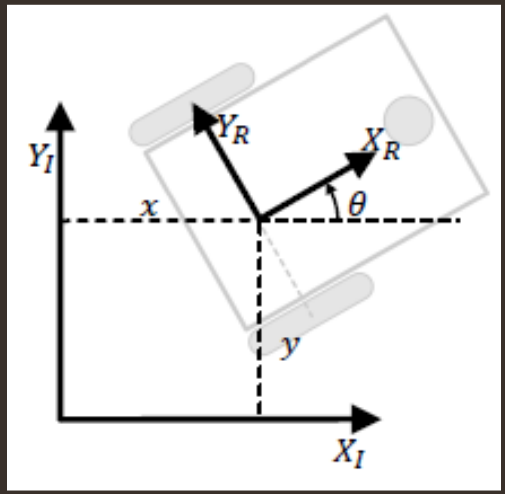


- **Motion control**

$$\begin{bmatrix} \dot{x} \\ \dot{y} \\ \dot{\theta} \end{bmatrix} = f(\dot{\varphi}_1 \cdots \dot{\varphi}_n, \theta, geometry)$$

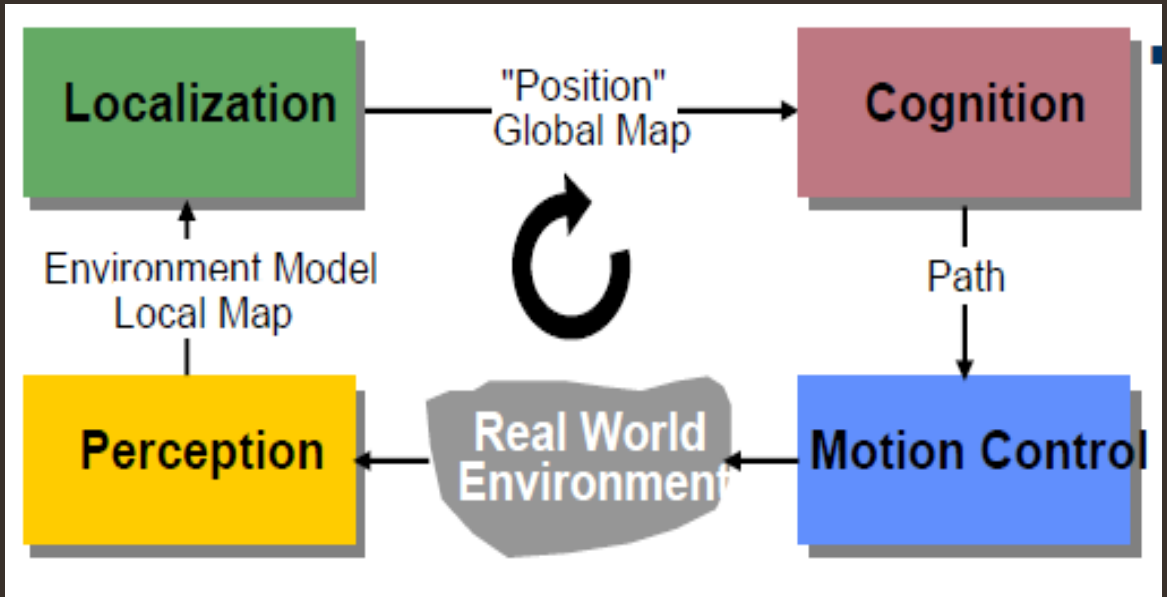
$$\begin{bmatrix} \dot{\varphi}_1 \\ \vdots \\ \dot{\varphi}_n \end{bmatrix} = f(\dot{x}, \dot{y}, \dot{\theta})$$

?



Control Architectures / Strategies

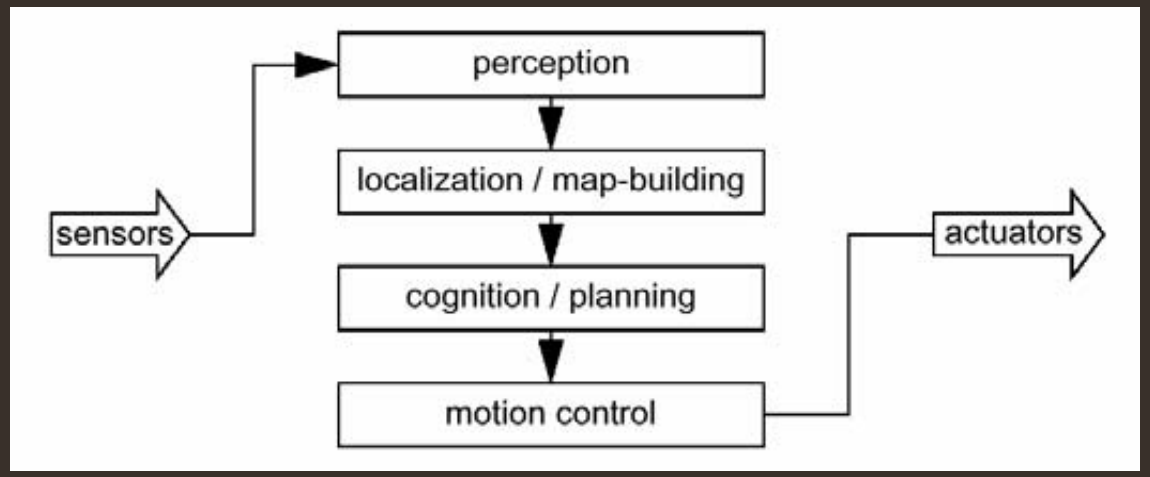
- Control Loop
  - dynamically changing
  - no compact model available
  - many sources of uncertainty

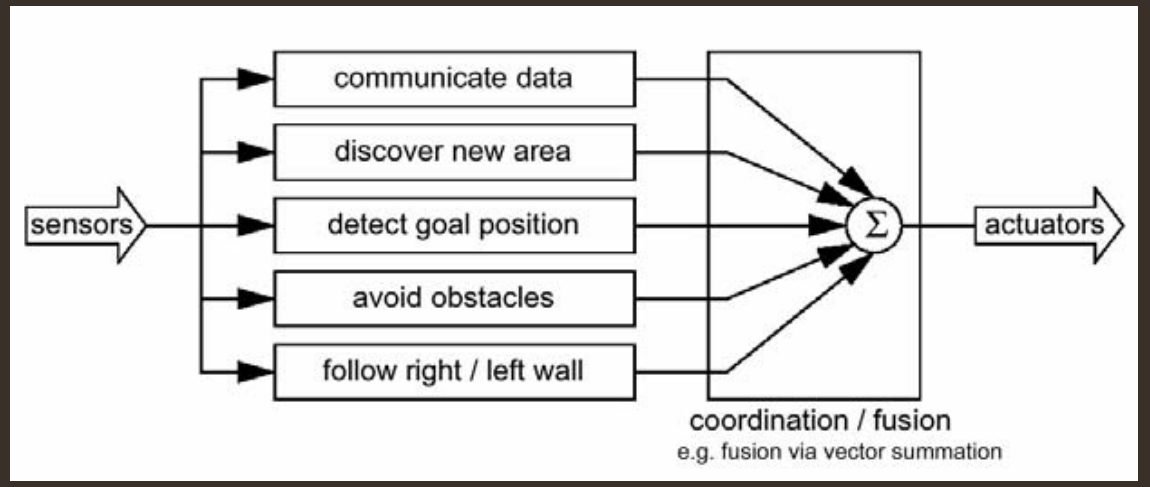


- Two Approaches
  - Classical AI
    - complete modeling
    - function based
    - horizontal decomposition
  - New AI
    - sparse or no modeling
    - behavior based
    - vertical decomposition
    - bottom up

Two Approaches

- Classical AI (planned based navigation)
  - complete modeling
  - function based
  - horizontal
  - decomposition
- New AI (behavior based navigation)
  - sparse or no modeling
  - behavior based
  - vertical decomposition
  - bottom up
  - Possible Solution
  - Combine Approaches





Planning Based Control

- Through perception and sensors fusion, a model of the “real” world is captured in memory.
- A goal is given and a plan is generated, assuming the “real” world is not changing.
- Then, the “plan” is executed, one (abstract) “operation” at a time.

Planning Based Control

- Example: A robot is equipped with a
- camera and two arms to perform an
- assembly task.
- Goal: put part A into part B
- World: where are A and B?
- Plan:
  - move left arm to A;
  - move right arm to B;
  - grab A; grab B;
  - move left and right arm closer;
  - assemble

  - What the camera sees is a world of "pixels".
  - What is "interesting" in the "real" world to be captured?
  - At what level of details should we represent the "real" world?
  - What if during plan execution, the "real" world changes? e.g., drop part A.

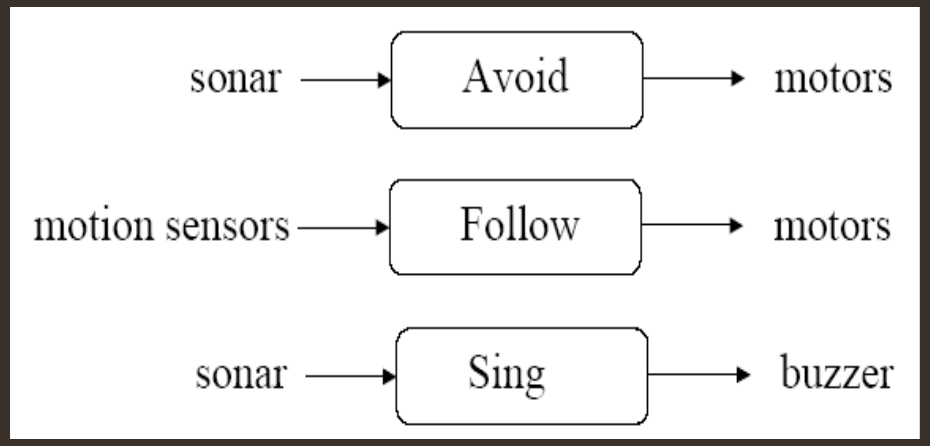
Behavior-Based Control

- Actions are connected to precepts via "behaviors".
- No internal model: The real world is our model.
- A robot reacts to changes and exhibits complex behaviors due to both internal and external interactions.

Behavior-Based Control

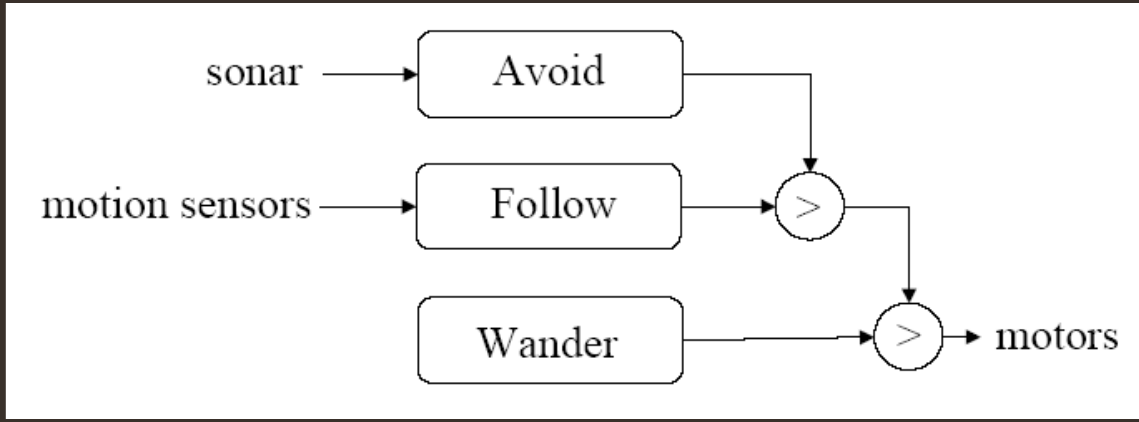
- **Control Structures**
  - A robot is equipped with many "simple" behaviors.
  - Each behavior defines its own sensor data and actions.
  - Interactions among the behaviors are resolved by coordination.
  - These behaviors are concurrent and independent; they react to changes instantly.
  - Different behaviors may share same sensors and/or actuators.
  - Competitive/cooperative actions are handled by careful coordination.
  - Behaviors may be added/deleted incrementally.

Example: A simple roaming mobile robot is equipped with the following behaviors:



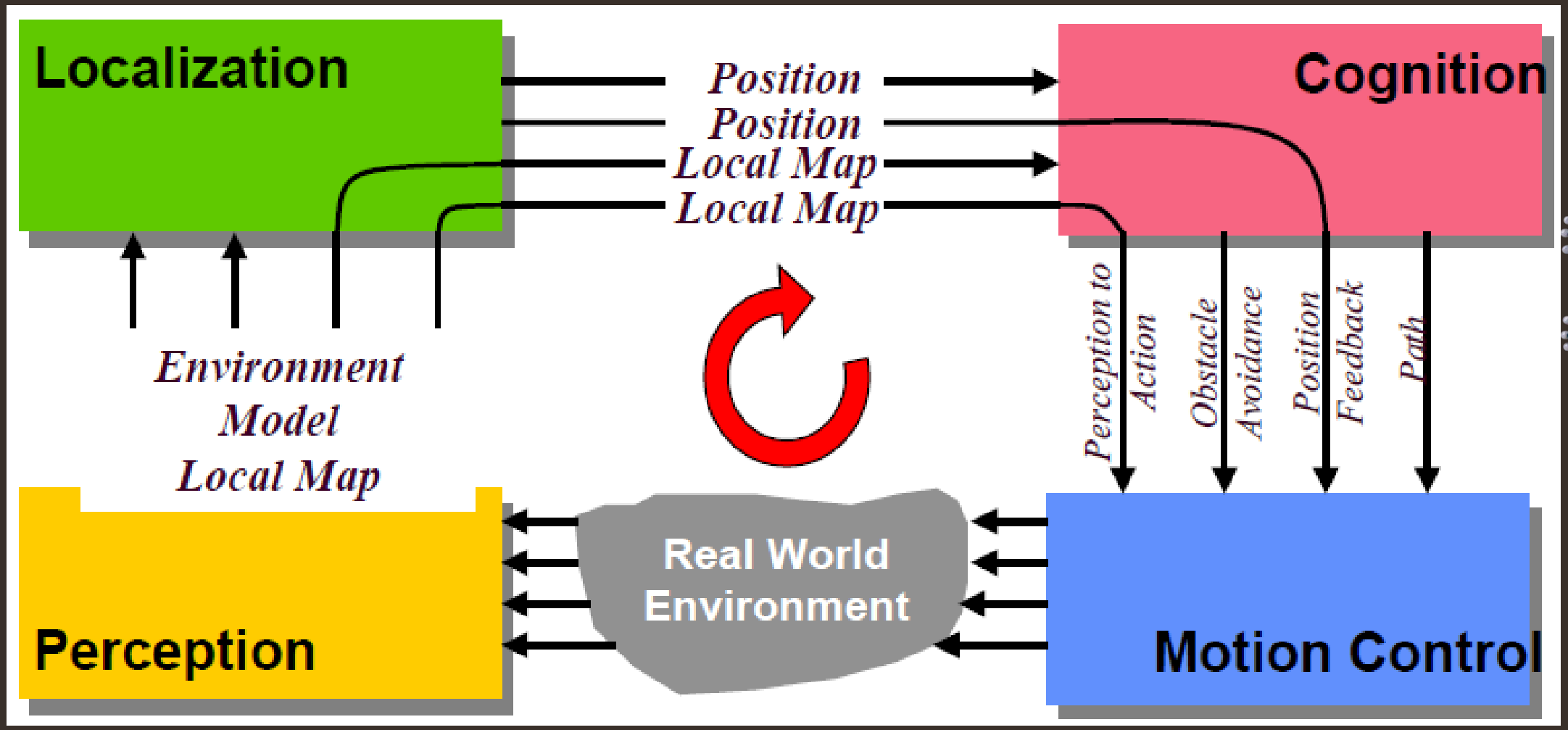
Subsumption Architecture

- Behavioral coordination can be based on a *fixed priority of suppression*.

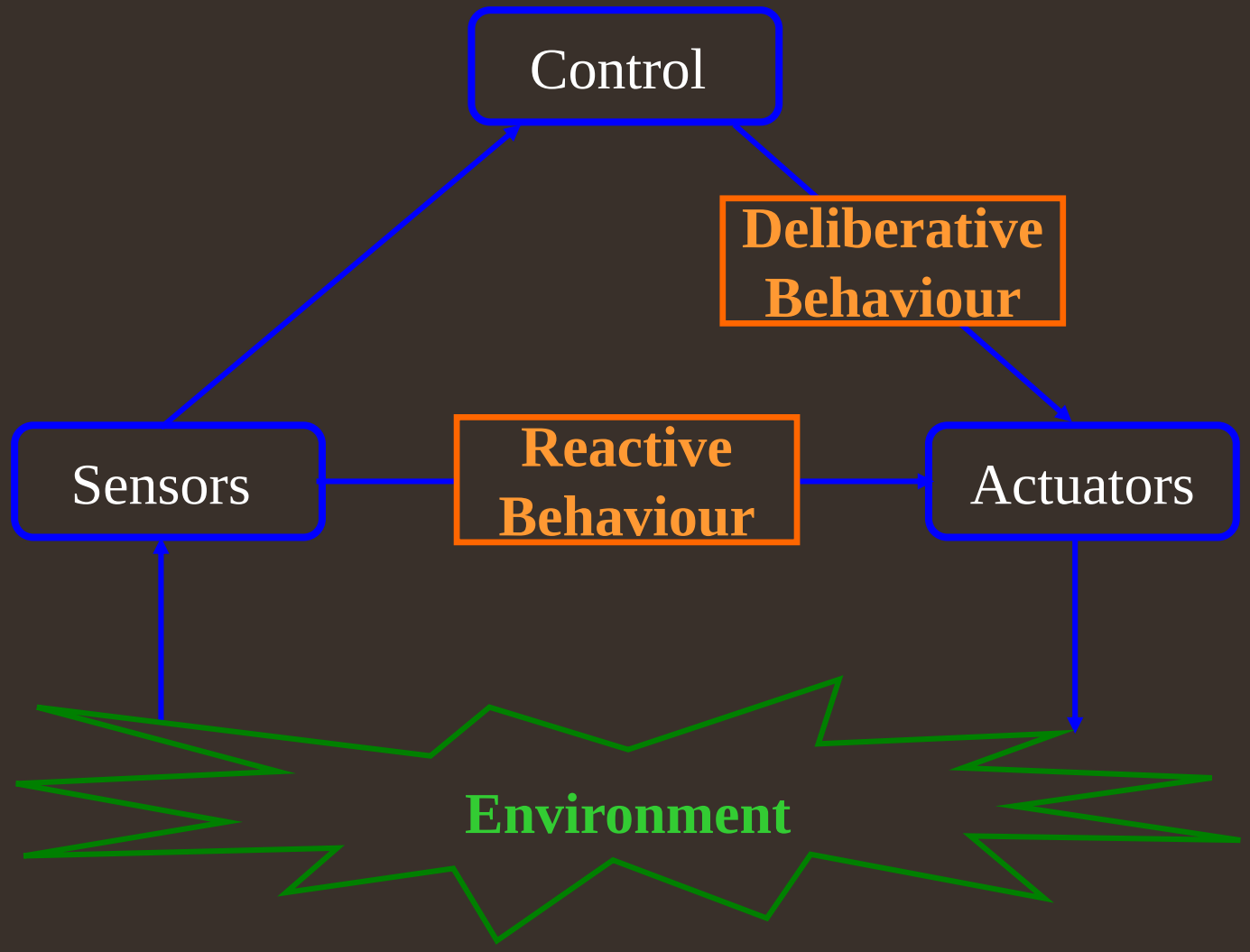


- Each behavior may generate more than one type of action.
- Multiple subsumption orderings may coexist at the same time.
- A behavior may be lower in one ordering, but be higher in an other.
- Each behavior's decision "cycle" time is independent.

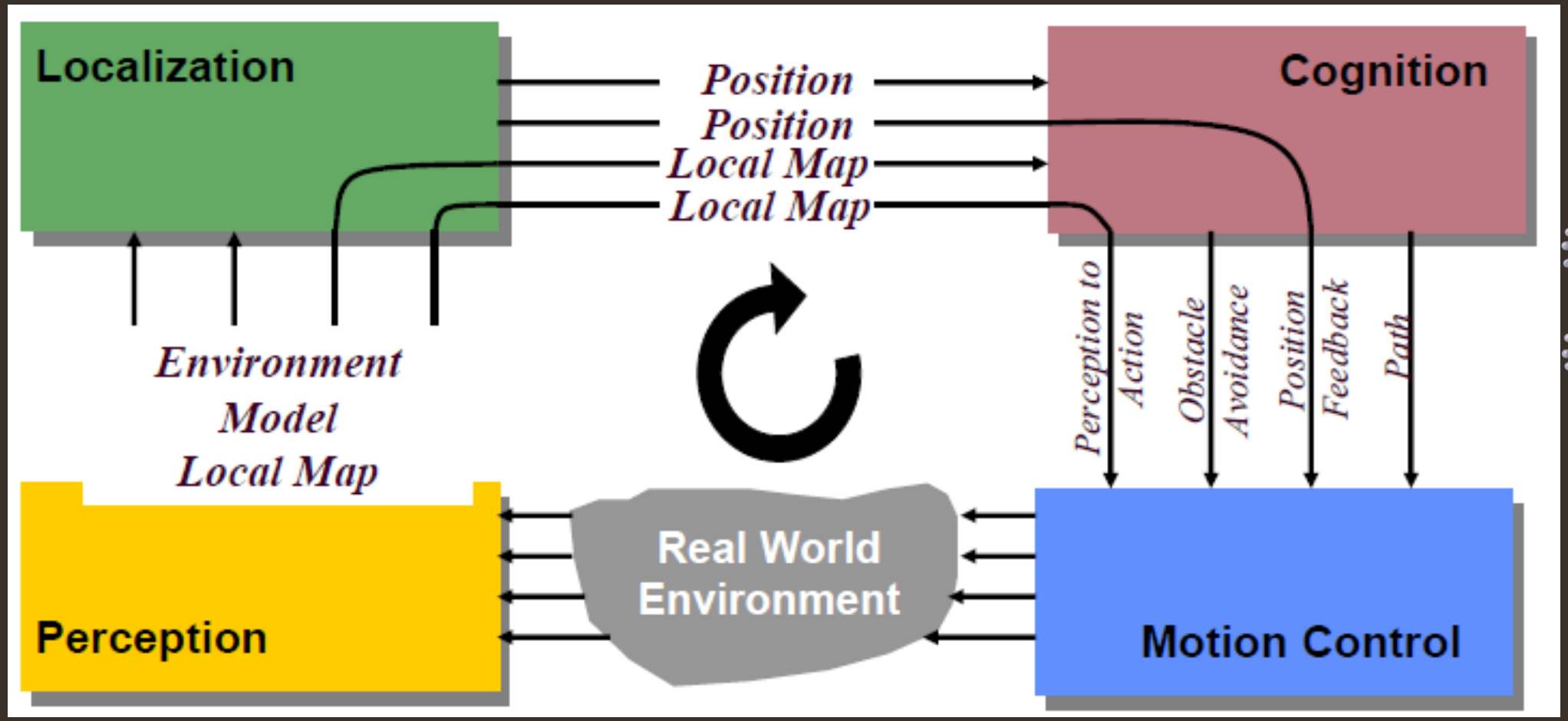
Mixed Approach



Mixed Approach



Mixed Approach



The Key for Autonomous Navigation

The Key for Autonomous Navigation

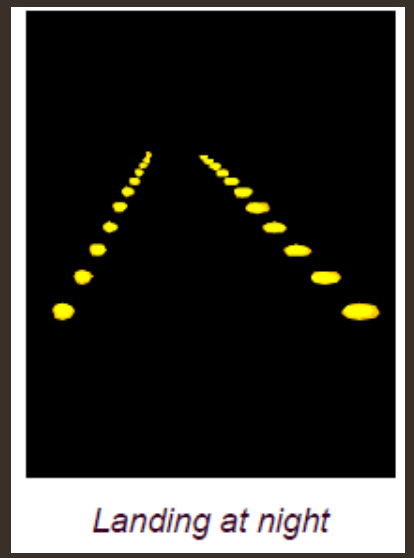
- Environment Representation
  - Continues Metric -> x,y,θ
  - Discrete Metric -> metric grid
  - Discrete Topological -> topological grid
- Environment Modeling
  - Raw sensor data, e.g. laser range data, grayscale images
    - large volume of data, low distinctiveness
    - makes use of all acquired information
  - Low level features, e.g. line other geometric features
    - medium volume of data, average distinctiveness
    - filters out the useful information, still ambiguities
  - High level features, e.g. doors, a car, the Eiffel tower
    - low volume of data, high distinctiveness
    - filters out the useful information, few/no ambiguities, not enough information.

Environment Representation and Modeling
How we do it!

Odometry

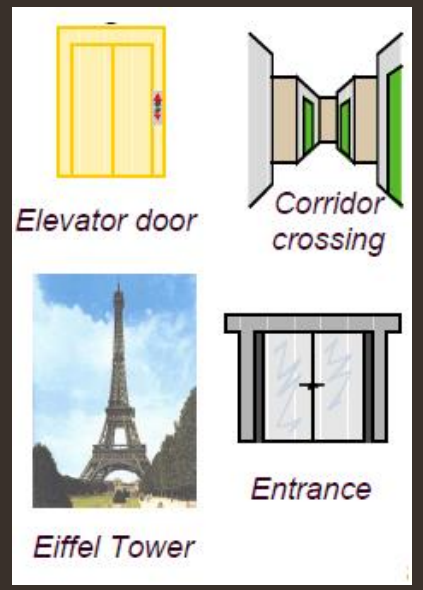
not applicable

Modified Environments



expensive, inflexible

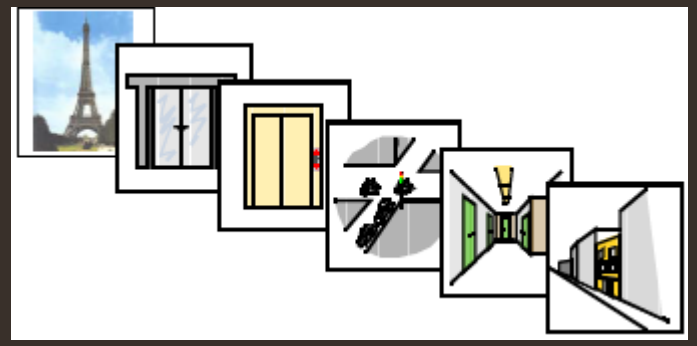
Feature-based Navigation



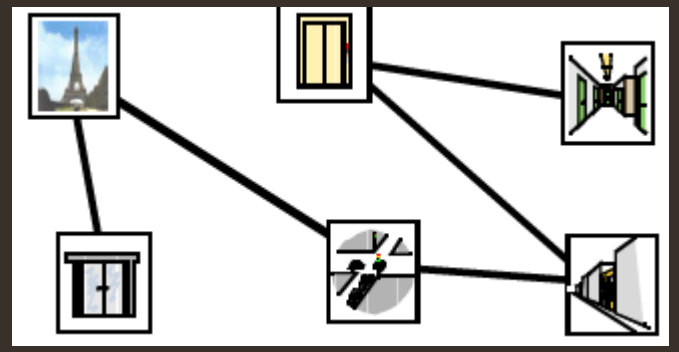
still a challenge for artificial systems

Environment Representation: The Map Categories

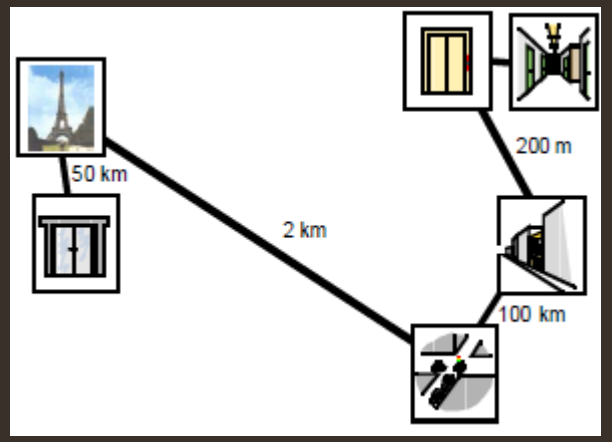
Recognizable Locations



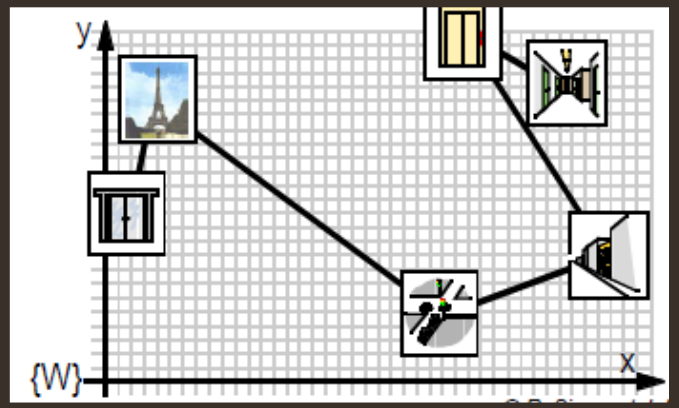
Topological Maps



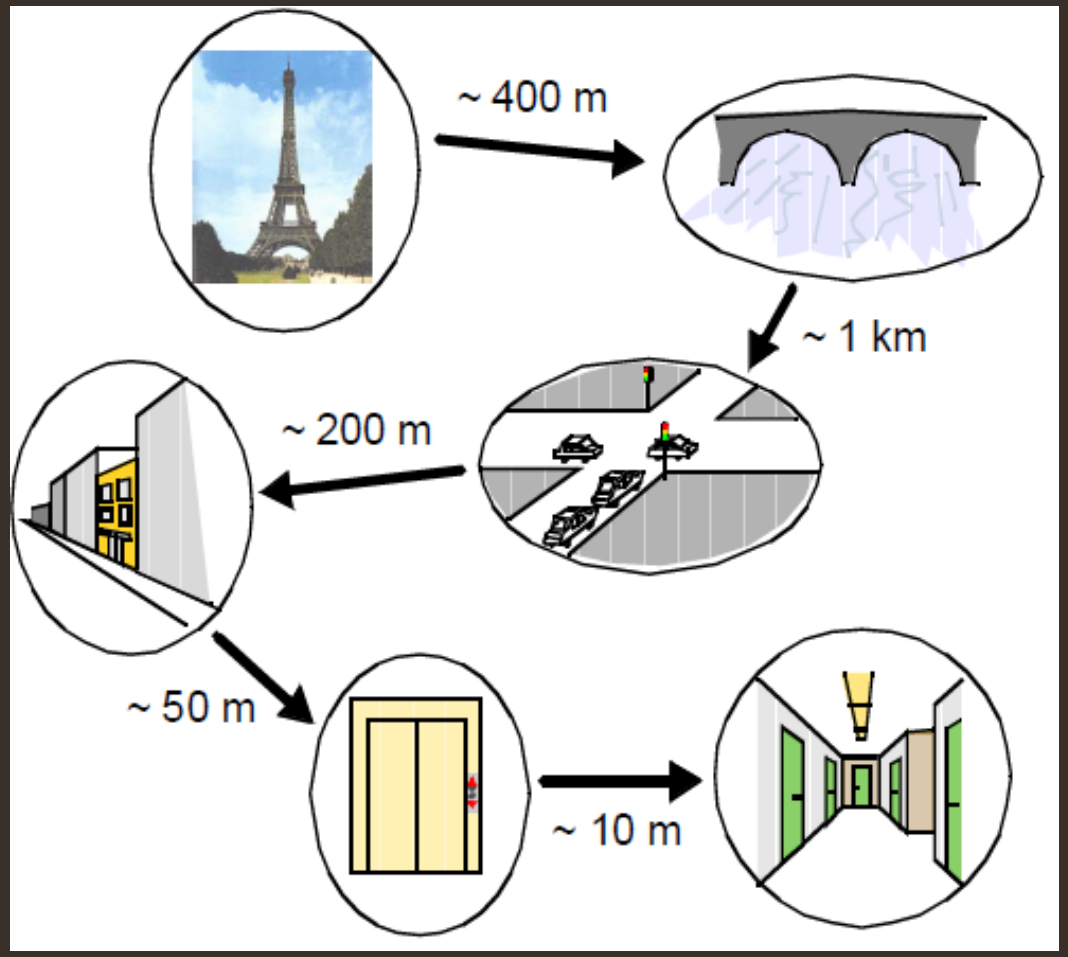
Metric Topological Maps



Fully Metric Maps
(continuous or discrete)

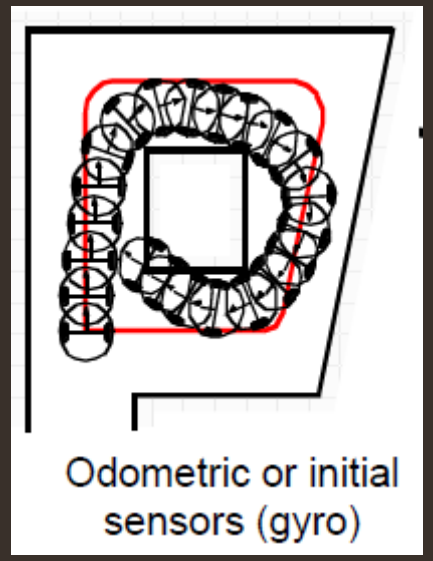


Human Navigation:
Topological with imprecise metric information



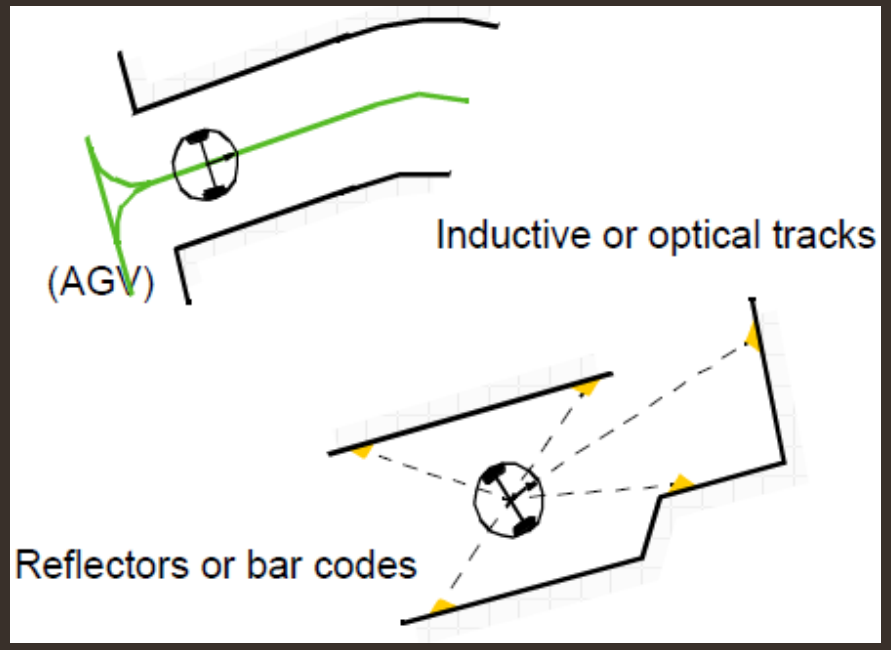
Methods for Navigation: Approaches with Limitations

Incrementally
(dead reckoning)



not applicable

Modifying the environments
(artificial landmarks / beacons)



expensive, inflexible

Summary

Summary

- Today we:
  - Defined what we mean by a robot
  - Investigated some of the areas in which robots are used
  - Defined the key components of a robot
  - Started to think about robot control mechanisms.

Questions?